

CLEARSKY OMEGA · SITE MAP EDITOR

Value Stack Funnel

How a project moves through ClearSky, how the value stack is built inside it, and where DividendVPP and Lightsmith come in. One sample site runs through both flows.

PREPARED FOR

DividendVPP (Molecule Systems) · Lightsmith Energy

PREPARED BY

ClearSky Energy Solutions

STATUS

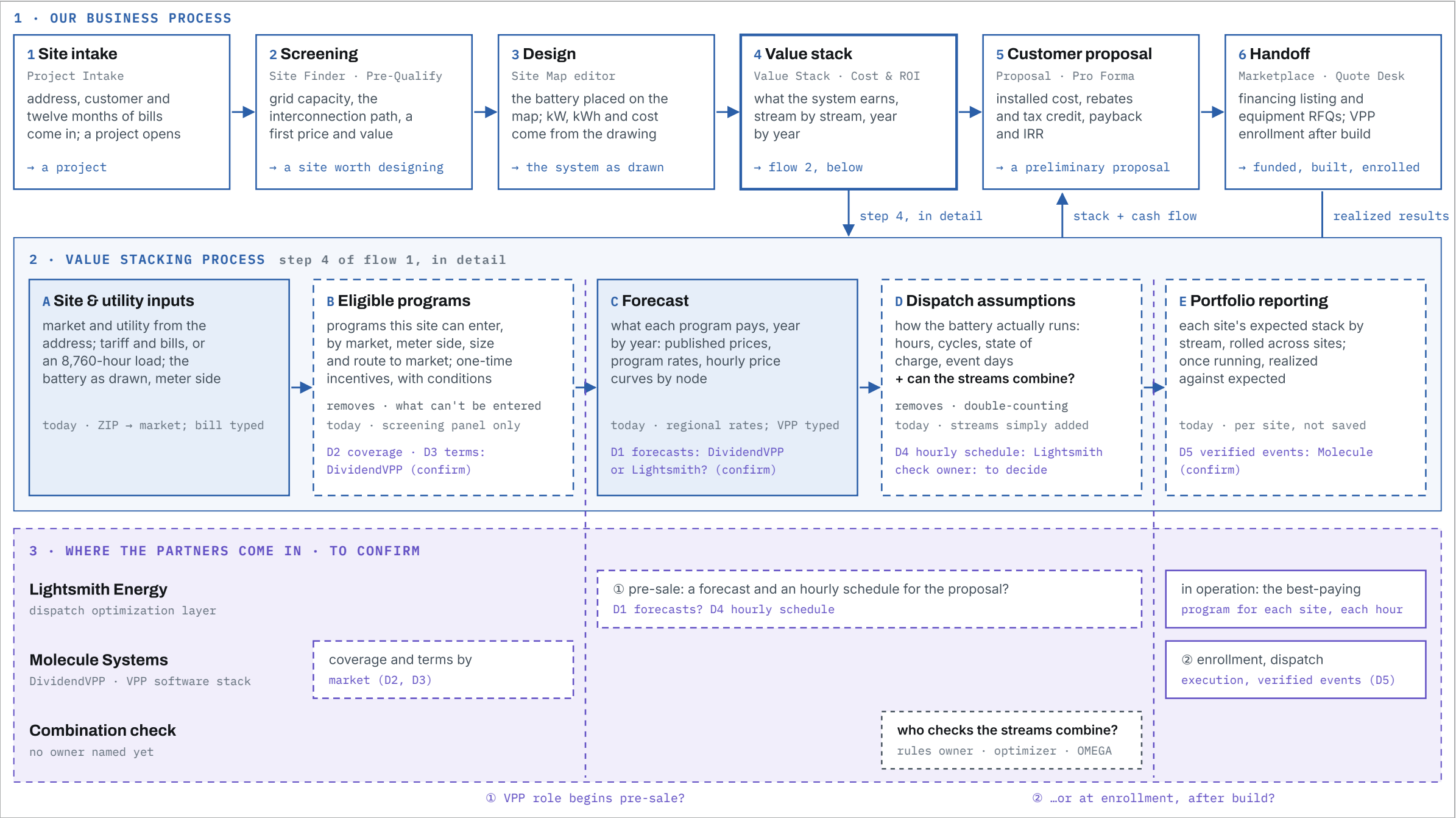
Discussion draft, rev B

DATE

28 September 2026

Two flows, one project. Flow 1 is our business process, from site intake and screening through the customer proposal and handoff. Flow 2 is step 4 of it: how the value stack is built, from site and utility inputs through eligible programs, the forecast and dispatch assumptions to portfolio reporting.

Today the number a customer sees rests on the inputs, a regional forecast and a ten-year cash flow. Program eligibility, dispatch and the check that revenue streams can be combined are **not yet applied to it**. That is where DividendVPP and Lightsmith come in, and where we need to agree who does what.



One sample site, through both flows

For walking the flows in the meeting. The site is illustrative; the Year-1 figures are what OMEGA's server-side screening model returns for it (the ComEd / PJM scenario), each labeled with where it comes from.

Sample site illustrative, for the walkthrough	
SITE C&I building, Chicago, IL; peak demand above 1 MW	UTILITY · MARKET ComEd · PJM, ComEd zone
METER SIDE Behind the meter	BATTERY 1,000 kW / 4,000 kWh, 4-hour
BILLS Twelve months, to replace the screening demand band	ROUTE TO MARKET To confirm in the meeting

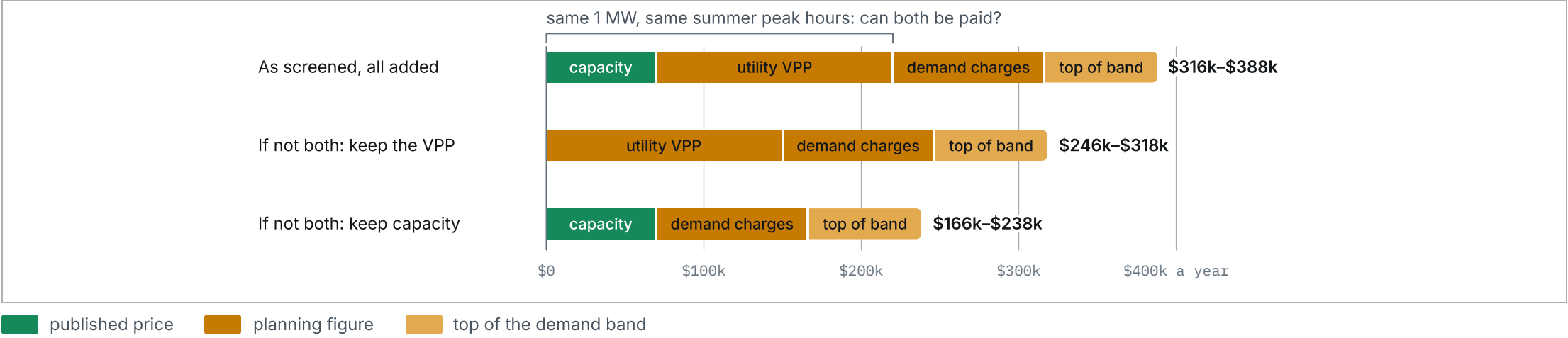


Figure 2. The sample site's Year-1 stack as OMEGA's screening model returns it, and what it becomes if PJM capacity and the utility VPP program can't both count the same 1 MW. Colors show where a figure comes from, not which stream it is.

FLOW 1, FOR THIS SITE

1 · Site intake The address, the customer and twelve months of bills come in through Project Intake; a project opens.	2 · Screening Site Finder prices it on the ComEd / PJM screening scenario; Grid Pre-Qualify checks the ComEd interconnection path.	3 · Design A 1 MW / 4 MWh battery is placed on the site map, behind the meter.	4 · Value stack Flow 2, below: the figures in the chart above.	5 · Customer proposal Installed cost, the ComEd rebate and the tax credit, payback and IRR, marked preliminary until verified.	6 · Handoff A financing marketplace listing and equipment RFQs; VPP enrollment after build.
---	--	--	--	--	---

FLOW 2, FOR THIS SITE

STEP	WHAT GOES IN	WHAT COMES OUT
A · Inputs	ComEd, PJM (ComEd zone); behind the meter; 1,000 kW for 4 hours. Until the bills are read, the model uses a demand-charge band of \$8–14 per kW-month.	The market, meter side and size that decide which programs exist for this site.
B · Eligible programs	PJM's 4-hour storage class; the utility VPP program (ComEd's is still before the ICC); the ComEd storage rebate; the federal tax credit.	PJM capacity, accredited at 59%; demand charges avoided; the utility VPP. One-time: the rebate at \$250/kWh, which needs a paired DG facility and Rate BESH, and the 30% tax credit, which needs prevailing wage. Route to market: to confirm.
C · Forecast	PJM's 2028/29 capacity auction at \$325/MW-day; the demand-charge band; a \$150/kW-year planning figure for the VPP.	\$69,989 capacity published · \$96,000–\$168,000 demand charges planning · \$150,000 utility VPP planning . Added: \$315,989–\$387,989 a year. One-time rebate: \$1,000,000.
D · Dispatch assumptions	All three streams want the battery in the same summer peak hours, and a 4-hour battery makes one full discharge a day.	The combination check. If the same 1 MW can't earn PJM capacity and the VPP program at once, the stack falls to \$246k–\$318k or \$166k–\$238k. Nothing checks this today.
E · Portfolio reporting	The site's expected stack by stream and its ten-year cash flow.	Rolled into the portfolio; once running, realized revenue per stream reported against this expectation.

Lightsmith and DividendVPP: what to confirm

What OMEGA has on record for each partner, and what we need each to confirm: the inputs they take, the outputs they return, where their VPP role begins, and who checks that revenue streams can actually be combined.

PARTNER	Lightsmith Energy DISPATCH OPTIMIZATION LAYER	Molecule Systems · DividendVPP VPP SOFTWARE STACK
ON RECORD IN OMEGA	Cross-program optimization, so each site is in the best-paying program available each hour.	AERA integrates the devices; each dispatch event is executed and verified by telemetry. A control and optimization layer, not a curtailment provider: a route to market sits underneath it.
TAKES FROM OMEGA CONFIRM	The asset (kW, kWh, round-trip efficiency, warranty cycles), an 8,760-hour load, the tariff, the node, and the eligible programs and enrollments. No customer name or address.	The commissioned asset and telemetry access, the programs chosen, and the customer's acceptance.
RETURNS TO OMEGA CONFIRM	An hourly program schedule and dispatch plan, forecast revenue per stream, and the value forgone where streams collide.	Coverage and terms by market (D2, D3); enrollment status; verified events and realized revenue by program (D5). A price forecast, if the "DividendVPP forecast" deck type is theirs (D1).
VPP ROLE BEGINS DECIDE	① Pre-sale, so the proposal carries its forecast and schedule; or ② only once the asset is enrolled.	At enrollment, after build. Coverage and terms shape the proposal before that.
TERMS CONFIRM	None on record.	Our tools assume different shares; we need one figure per market and program.

Who checks that revenue streams can actually be combined? Today, no one. OMEGA's revenue model adds the streams, and no program's rules on combining them are held anywhere in OMEGA.	
On the sample site , PJM capacity (\$69,989) and the utility VPP (\$150,000) would both count the same 1 MW in the same summer peak hours, and managing the demand charge needs the battery at the monthly peak as well. Whether the same kW can be paid twice is a program rule, not a modeling choice. The answer moves this site's stack by \$70k to \$150k a year .	The market participant of record, the aggregator or program operator, owns each program's combination rules, because the program holds them to it. Lightsmith's optimizer enforces the rules hour by hour, choosing one program for an hour where two can't combine. OMEGA applies the rules before a stack reaches a proposal: a stream the rules exclude isn't added, and the proposal says which combination it assumes.
Proposed: the participant of record owns the rules, Lightsmith enforces them in dispatch, and OMEGA applies them in the stack. To decide in the meeting: the named owner, and where the rules table lives.	

What each point would send, and in what shape

POINT	FROM CONFIRM	WHAT	SHAPE OMEGA NEEDS	WHERE IT LANDS
D1 Rates & forecasts	DividendVPP or Lightsmith	Rates for the programs you enroll assets in; hourly price forecasts	Rate, unit (per kW-year, per kW-summer, per event), program, utility, source and effective date. Forecasts as 8,760 hourly rows per product per settlement point, unit declared (\$/MWh or \$/MW-h), negative prices kept, a vintage label.	Step C. Fills the VPP line, typed in today, and the program and capacity rates, each with its source. A forecast becomes a price deck locked to its vintage.
D2 Coverage	DividendVPP, with the route to market	Markets, utilities and programs you can enroll into	Per program: market, utility territory, behind or in front of the meter, minimum kW, telemetry, season.	Step B. Which streams the route reaches, resolved by utility territory rather than by state.
D3 Terms	DividendVPP, with the route to market	Platform fee or revenue share	Per market and program, with any minimums.	Step B. What stays with the owner on this route.
D4 Dispatch	Lightsmith	A co-optimized hourly schedule	Request: kW, kWh, round-trip efficiency, warranty cycles, 8,760-hour load, tariff, node, and no customer name or address. Response: hourly charge, discharge and state of charge; revenue per stream; value forgone.	Step D. An hourly run on one state of charge instead of streams added together: the number a lender funds.
D5 Verification	Molecule (telemetry)	A figure you stand behind; verified events once running	Verification ID, date, the assumptions behind it, and how long it holds; each event's telemetry once enrolled.	Steps D and E. Replaces the "~ Estimate" badge and travels with the project to reports, the pro forma and the financing listing.

Enrollment would follow the partner route OMEGA already runs for equipment vendors. The vendor sees an anonymous opportunity (state, utility, size, stage), and the customer's identity is released only when the customer accepts.

Inside the editor, click by click

Steps 3 to 5 of flow 1 as the screens a developer, installer or OEM opens, in the order they open them.

STEP 1

Put the battery on the map

Build › BESS Build › BESS Sizer › Solar + Storage

The value stack reads kW and kWh off the placed fleet, then the configured list. No battery drawn, no stack.

flow 2 › A

STEP 2 › OPTIONAL

Screen the market

Analyze › Screening › Value Stack

ZIP to market, behind or in front of the meter, and who monetizes it: self, a utility program, an aggregator or Molecule Systems. A screening view; its figures stay in the panel.

flow 2 › A and B, screening only

STEP 3

Estimate the streams

Estimate › Cost & ROI › Construction Cost › Revenue

ZIP (prefilled), the customer's monthly bill and facility type, plus a regulation price in PJM, then Estimate Value Stack. A front-of-meter project is turned away here.

flow 2 › C

STEP 4

Apply and fill the gaps

Revenue › Apply to Revenue Streams

Arbitrage, peak demand, demand response and PJM regulation fill in with an "~ Estimate" badge. Virtual power plant, capacity and resilience are typed in by hand.

flow 2 › C

STEP 5

Read the return

Cost & ROI › ROI › results rail

Payback, year-one return, and a ten-year NPV and IRR, with revenue escalated at a fixed rate.

flow 1 › step 5

STEP 6

Present and hand off

Export Analysis › Output › Marketplace

The analysis export prints an Estimated Year-1 Value Stack marked preliminary. Push to Marketplace carries the size and asking price, and Apply for Financing the project itself; neither carries the stack.

flow 1 › step 6

Market by market

What a battery can be paid for in each market, and how the number that reaches the customer is valued today. Each amber, blue or outlined cell is a place partner data would change the answer.

Estimateregional planning figure

Typed inno rate on file; the user enters one

Not valuedthe market pays for it; the model doesn't count it

Comingbuilt, not open yet

No marketnothing to earn here

MARKET	CAPACITY & DEMAND RESPONSE	ANCILLARY SERVICES	VPP PROGRAM	ENERGY ARBITRAGE	WHAT PARTNER DATA WOULD SHARPEN
PJM Mid-Atlantic, Ohio Valley, ComEd	Estimateone band per kW-year	Estimateregulation at an editable clearing price	Typed in	Estimateretail time-of-use spread	Rates for the utility VPP and demand-response programs you enroll into; node energy and regulation forecasts
ERCOT Texas, energy-only	No market	Not valuedComingthe ERCOT price deck takes all five services	Typed in	EstimateComingretail spread today; nodal via the price deck	Hourly DA, RT, Reg-Up, Reg-Down, RRS, ECRS and Non-Spin at the settlement point: the seven series the price deck already takes
CAISO California	Estimateresource adequacy is bilateral	Not valued	Typed in	Estimate	Program rates by utility; RA values; ancillary forecasts
NYISO New York	EstimateICAP varies by zone	Not valued	Typed in	Estimate	ICAP by zone; VDER and program rates; ancillary forecasts
ISO-NE New England	Estimate	Not valued	Typed in	Estimate	Program rates by state and utility, with event history; ancillary forecasts
MISO Central US	Estimate	Not valued	Typed inaggregated DR opening state by state	Estimate	Which utilities allow aggregated demand response, and what it pays
SPP Southern Plains	Estimatea default band	Not valued	Typed in	Estimate	Program availability; ancillary forecasts
Outside an ISO Southeast, West	Estimatea default band	No market	Typed in	Estimateutility time-of-use spread	Utility programs you enroll assets into

Bill-side value, demand charges and time-of-use savings, comes from the customer's own monthly bill in every market.

The screening panel (Analyze › Value Stack) goes further in places: PJM capacity at the published auction price, ancillary services for front-of-meter assets, and named programs such as ConnectedSolutions, SGIP and NYSERDA. Its figures stay in that panel today.

The market is resolved from the site's state. Where a state straddles two markets (Illinois, Indiana, Michigan, Kentucky, North Carolina), the utility territory or pricing node should decide, which is part of what D2 coverage settles.

Rules the editor already holds itself to

Anything a partner sends would be held to the same rules.

The stack sizes off the drawing. kW and kWh come from the battery on the site map or its configured list, never from a number typed into the revenue form. No battery, no stack.
A missing rate is shown, not hidden. A stream the market doesn't have is marked n/a, one with no rate on file stays at zero until someone enters it, and a utility the rate table doesn't know is flagged "default, verify".
Estimates say so. Every applied stream carries an "~ Estimate" badge, every figure in the screening panel shows its arithmetic, and exports call the stack preliminary until it is verified.
Price imports ask for units. \$/MWh and \$/kWh differ by a thousand, so the price-deck importer asks instead of guessing, keeps negative prices (the hours a battery earns by charging) and locks a deck to its vintage so a case already sent cannot re-baseline.
A merchant asset is not given a retail answer. The retail estimator turns away a front-of-meter project instead of valuing it with retail rates it will never see.

Where things stand

IN USE TODAY - Estimate › Cost & ROI › Revenue: the estimator, applied to revenue streams, the ROI and the results rail - Analyze › Value Stack: a screening view by market and route to market, with Molecule Systems as one route - An analysis export with an Estimated Year-1 Value Stack, marked preliminary - Owner reporting and the O&M console: expected against actual energy and availability per site, not revenue by stream	BUILT, NOT OPEN YET - Price Decks: nodal energy and ancillary series by settlement point, with a "DividendVPP forecast" provider tag - An allocation engine that spends 8,760 hours and the warranty's cycles across streams, and a lifecycle model with fade and augmentation - The VDC Exchange's data rules, with Molecule and Lightsmith set up as operators; the exchange page itself is not built	PROPOSED HERE - D1 to D5, server to server between OMEGA's /api and the partners' APIs - A named owner for the combination check, and one rules table the stack and the dispatch both read - The verified stack carried to reports, the pro forma and the financing marketplace - To start: a sample forecast export, the coverage list, terms by market, Lightsmith's input and output spec, and dispatch API documentation
---	--	---